

0009 — MenthorQ Gamma Data × MC Edge — 2026-07-04

****Series:**** MC Setup Research Notes · Note 0009 ****Confidence:**** Medium — 81 trading days of MenthorQ data (Mar 6 – Jul 2, 2026), 374 filled trades / 208 Stack-v2; full-cost sims on tick data; every hypothesis pre-registered before outcomes were seen; but the sample is one 4-month regime and the archive scalars have a timestamp caveat (§7). ****TL;DR:**** We bought 3 months of MenthorQ dealer-gamma data (Call Resistance, Put Support, HVL, gamma walls, GEX strikes, blind spots, net GEX, QScores) and threw four rounds of pre-registered tests at it. ****The levels are not intraday support/resistance**** — they bounce no more often than random prices, and neither do prior-day H/L/C, VWAP, value areas, or the IB (nothing does, at the 5M scale, in this window). *****Don't trade into a level" is refuted**** — those trades did better. The two things that survived: ****negative-gamma days realize ~1.18× the implied expected move vs 0.96× on positive days**** (gamma is a volatility amplitude dial, not a directional or level tool), and ****trading into a ≥3-level cluster within 1R was flat-to-negative while everything else earned +0.19R**** (top watch item). ****No change to the frozen Stack v2 spec.**** The data's future value is sizing/vol-conditioning, re-testable at 6–12 months.

1. The setup (so this note stands alone)

The book being filtered. MC ("MyMicroChannel") 5-minute breakout signals on ES, filtered by the S53 frozen "Stack v2" spec: skip counter-IB-break signals, skip days after a prior trend day ($>1.6 \times \text{ADR}_{14}$), skip signals at/after 14:00 CT. Exits: 3R target, ratchet to break-even at +1R, flat by end of session. Costs everywhere in this note: 1-tick entry slip, 1-tick exit slip, 1-tick stop offset, \$4.36 round-turn commission, ES \$50/pt. $R = |\text{entry} - \text{stop}|$. Over 2021–2026 this book backtests $\approx +0.16R/\text{trade}$; in this study's 4-month window it runs $+0.157R \pm 0.166$ ($n=204\text{--}208$ filled stacked trades, consistent with the long-run number).

The data being tested. MenthorQ publishes, per trading day, option-positioning levels for ES derived from dealer gamma exposure: **Call Resistance** (largest call-gamma strike, "dealers sell into strength there"), **Put Support** (largest put-gamma strike, "dealers buy dips there"), **HVL** (the gamma-flip level: above = positive dealer gamma = vol-dampening, below = negative = vol-amplifying), 0DTE variants, **Gamma Wall 0DTE**, ten **GEX strikes** (25-pt option strikes with big exposure), ten **Blind Spots (BL)** (proprietary levels), plus daily scalars: net/total/expiring GEX, P/C ratios, IV30, 1-day expected move %, a Positive/Negative **gamma condition** flag, and four 0–5 **QScores** (option, momentum, volatility, seasonality). We back-filled **82 daily snapshots (2026-03-06 → 2026-07-02)** via their API into data/menthorq/menthorq_levels.csv (81 usable after joining to bar data).

Price-space alignment (matters!). Our bars/signals/ticks are back-adjusted continuous; MenthorQ levels are front-contract prices. Measured per-contract offsets from actual contract bars: **ESH6 +111.00**, **ESM6 +61.25**, **ESU6 +0.00** (each constant within its period, matching NT rollover records). All levels were shifted into continuous space before any distance was computed. Skipping this step would mis-place every level by 50–110 points.

2. The question

Does dealer-gamma positioning data improve the MC entry edge — as level S/R, as an entry filter (don't trade into levels / trade break-and-retests / avoid clusters), or as a day-regime dial? And a method question underneath: at what unit (touch, trade, day) does this data even have a chance of showing value?

3. How we tested it

All hypotheses were **pre-registered** (direction declared before looking at outcomes), coarse bins only, all cuts reported. Across the study we ran ~50 cuts — expect ~4–5 spuriously "significant" rows by chance; single-cut stars are treated accordingly.

1. **Stage 0 — data hygiene:** roll-offset alignment; are the fields causal (pre-market) or EOD-stamped? 2. **Stage 1 — level S/R touch test:** 5M touch → bounce/break within 6 bars (3-pt threshold), ~2,000 real touches vs matched random controls. 3. **Stage 2 — trade-level pre-registered tests (T1–T5):** headwind/tailwind, gamma regime, QScores, IV/expected-move, the high-volneg-gamma combo — on the stacked book with frozen exec. 4. **Follow-up A (user challenge: "main levels must have merit"):** day-range **containment** vs a distance-matched benchmark; first-touch-to-EOD outcomes (T-A/T-B/T-C). 5. **Follow-up B (user rules):** U1 "never trade into a main level" ($\leq 1R$), U2 break-and-retest, U3 clusters, U4 expanded universe (prior-day H/L/C, VA, IB, VWAP) as the interpretive anchor. 6. **Follow-up C (day as the unit, $n=81$):** D1 gamma vs realized/implied range, D2 gamma vs trendiness, D3 corridor geometry, D4 expiring-GEX pinning, D5 book day-P&L by regime.

4. Results

4.1 Stage 0 — hygiene findings that gate everything else

- 1d_max/1d_min are an **expected-move band** (width $\equiv 2 \times$ expected move, ratio 1.00; realized range fully inside 78% of days) — a forecast, causal, usable.
- **! The archive rows are EOD-stamped:** distance_to_hvl_% matches the SAME-day close (median error 0.05pp) not the

prior close (0.46pp). The OI-derived levels are computed from prior-day close data and static intraday (causal), but the archived gamma_condition and QScores may embed same-day information. Every regime/score test therefore has a **causal prev-day variant**, and live use requires **pre-market capture** (the app's Refresh button) — do not trust backfilled same-day scalars.

4.2 Stage 1 — are the levels intraday S/R? ****No.****

Touch = 5M bar crosses the level; bounce = ≥3-pt rejection without a 3-pt penetration within 6 bars; controls = random in-range prices (or other 25-pt strikes for GEX), ≥5 pts from any real level, seeded.

level group	touches	bounce%	break%	ctrl bounce%	bounce diff 95% CI
KEY (CallRes/PutSup/HVL/GW)	611	21.8%	78.1%	23.9%	[-5.4pp, +1.4pp]
GEX strikes	747	24.1%	75.5%	20.4% (n=103)	[-3.5pp, +12.1pp]
BL blind spots	647	22.3%	77.1%	24.2%	[-5.6pp, +1.8pp]

4.3 Stage 2 — trade-level pre-registered tests (stacked book, n=208)

Baseline in window: **+0.150R ±0.163, PF 1.29, WR 37%** (all-signals +0.061R ±0.109, n=374). Registered hypotheses and outcomes:

hypothesis (registered direction)	stacked result	verdict
T1 headwind: opposing major level inside the 3R path → worse	+0.149 (headwind, n=115) vs +0.151 (clear air, n=93)	null
T1 distance-to-opposing tertiles	near +0.108 / mid +0.386 [+] / far +0.036 — non-monotone	noise shape
T2 negative gamma → better (momentum book)	+0.148 vs +0.151 (as-published); causal prev-day -0.027 vs +0.244 [+] — wrong sign	null / refuted
T2b above vs below HVL	+0.147 vs +0.156	null
T3 vol_score low ≤1 / mid / high ≥4	+0.182 / +0.063 / +0.183	null
T3 momentum, option scores (coarse)	all CIs span 0	null
T3 seasonality_score ≥2	-0.419 ±0.232 (n=32, PF 0.21, WR 19%) ; all-signals scope -0.313 ±0.204 (n=49)	only consistent negative pocket — but 12 calendar days, 1 of ~35 cuts → watch item
T4 net GEX sign	≡ gamma condition split — null	null
T4 IV30 / exp-move tertiles	low -0.04 / mid +0.33 [+] / high +0.21 — non-monotone	suggestive "edge needs some vol", nothing more
T5 user combo: high-vol(≥3) AND negative gamma → better	+0.053 (n=50) vs rest +0.181 ; causal +0.087 vs +0.171	refuted (slightly underperformed)

4.4 Follow-up A — "the main levels must have merit" (containment & first touch)

The observation is real; the explanation isn't gamma. Containment = the day's extreme stays inside the level. Benchmark = empirical CDF of the day's (high-open)/expected-move evaluated at each level's distance — i.e., "how often would ANY price that far away contain the day?"

level	days usable	contained	benchmark	excess	boot 95% CI
Call Resistance	80	95%	86%	+9.4pp	[+4.6, +13.9] [+]
Call Res 0DTE	74	82%	72%	+10.1pp	[+2.6, +17.0] [+]
1d_max (pure IV band, ref)	79	94%	78%	+15.5pp	[+9.9, +21.5] [+]
Put Support	77	94%	90%	+3.4pp	[-2.5, +9.2]
Put Sup 0DTE	71	79%	77%	+2.3pp	[-7.7, +11.2]
1d_min (pure IV band, ref)	75	92%	78%	+14.0pp	[+7.6, +19.6] [+]

Call Resistance does cap the day 95% of the time — but a strike-free IV band caps it better. The containment is **distance + implied vol**, not gamma placement. First-touch-to-EOD stats are untestable at the main levels' event rate (Call Res touched 4 days of 81, Put Support 5); when Put Support was touched at the 6-bar scale it broke 89% of the time. Gamma Wall / HVL first-touch rejection depths beat matched controls in point estimate (48.0 vs 18.5 pts median) but every CI spans zero.

4.5 Follow-up B — user rules + the expanded level universe

U4 anchor (read this first): the same touch machinery applied to the levels "everyone knows work":

family	touches	bounce%	break%	bounce diff vs 3,263 random-control touches
MQ main	407	20.1%	79.6%	[-5.0pp, +3.5pp]
MQ other (BL+GEX)	1,394	23.2%	76.3%	[-0.4pp, +5.0pp]
Prior day H/L/C	426	24.4%	75.6%	[-0.8pp, +7.6pp]
Prior day VA (POC/VAH/VAL)	374	21.7%	78.3%	[-3.8pp, +5.0pp]
IB high/low (post-IB)	315	21.3%	78.1%	[-4.4pp, +4.9pp]
VWAP (developing)	448	22.5%	77.2%	[-2.5pp, +5.5pp]
random control	3,263	21.0%	78.5%	—

Nothing — not prior-day high/low, not VWAP, not value areas — bounces more than random at the 5M/30-min scale in this

window. The MenthorQ null is not special; visual level-respect at this scale is mostly base rate + confirmation bias (or an effect smaller than 81 days can detect).

U1 "never trade into a main level" (registered: worse): REFUTED.

cut (stacked)	n	ExpR ±CI	PF	net\$
INTO MQ main level (≤1R)	48	+0.262 ±0.362	1.32	\$7,591
not into MQ main	156	+0.124 ±0.186	1.29	\$18,245
INTO any level (MQ or structural)	139	+0.209 ±0.211	1.43	\$28,494
clear of ALL levels (≥1R air)	65	+0.044 ±0.260	0.88	-\$2,658

Trading into levels did fine; clear air was the worst bucket. Skipping into-level trades would have cost money.

U2 break-and-retest: untestable — 16 qualifying trades all-scope, 0 in the stacked subset (main levels rarely break + retest + align with a signal). No conclusion.

U3 clusters (registered: worse): the only registered-direction hit of the study. Cluster = ≥3 levels (MQ + structural) chained within 4 pts; 175 zones over 81 days.

cut	stacked	all signals
INTO cluster zone (≤1R)	+0.005 ±0.386, PF 0.76 (n=33)	-0.100 ±0.237, PF 0.64 (n=63)
not into cluster	+0.186 ±0.183 [+], PF 1.43 (n=171)	+0.097 ±0.123, PF 1.34 (n=307)
cluster touch test	bounce 26.5% vs isolated 22.0%, CI [-2.2, +10.7]	leans right, n.s.

Registered direction, consistent across both scopes, coherent mechanism (confluence). Small n → **top watch item, not a rule.**

4.6 Follow-up C — the day as the unit (n=81): where gamma finally shows up

test (registered direction)	groups	diff	95% CI		
D1 realized range / expected move: NEG vs pos gamma (reg: NEG higher)	1.177 (n=33) vs 0.960 (n=48)	+0.217	[+0.032, +0.412] [+]		
D1 causal (prev-day label)	1.131 vs 0.999	+0.132	[-0.062, +0.342]		
D2 trend efficiency \	C-O\	/(H-L): NEG vs pos (reg: NEG higher)	0.472 vs 0.458	+0.014	[-0.105, +0.131]
D3 corridor width vs range/EM (Spearman)	r = -0.044		null		
D3 open position in corridor vs drift	r = -0.122		null		
D4 expiring-GEX share vs afternoon compression	r = +0.040		null		
D5 book day-P&L: NEG vs pos gamma	+\$381 vs +\$276/day		[-1349, +1723] null		
D5 book day-P&L: days realizing > vs < implied move	+\$1,006 vs -\$386/day	+\$1,392	[+46, +2806] [+] (diagnostic)		

D1 is the first MenthorQ claim to pass a test on our data — negative-gamma days overshoot the implied expected move, positive-gamma days undershoot it. Caveat: the as-published label is EOD-stamped and partly derived from where spot ended vs HVL (a big day can label itself negative); the causal prev-day version points the same way but isn't significant. **D2 shows the amplification is amplitude, not direction.** D5's significant row is diagnostic, not causal (you don't know the day's range in advance): it proves the stacked book is **long realized volatility** — the exact quantity a clean morning gamma/vol signal would forecast.

5. Why it works / fails (the synthesis)

Gamma has been used profitably for decades — but look at how: market-makers' gamma scalping (harvesting realized vol), vol-risk-premium selling in pinned regimes, OPEX pin trades, and modern funds using dealer-positioning as a **risk dial** (widen stops / cut size in negative gamma). Every proven use is a **volatility or sizing use**. "Buy the bounce off the gamma level intraday" — the retail rendering — is the one use this data consistently fails to support. Our results reproduce that division precisely: the amplitude effect is real (D1), the direction effect absent (D2), the level-bounce effect absent for every level family including classical ones (U4), the containment "merit" of the main levels is explained by implied vol alone (T-A), and our book's P&L is literally a bet on realized vol (D5). The data's value to us, if any, is as a **morning vol/sizing input** — which the current archive can't deliver cleanly (EOD stamps) but our own pre-market capture will.

6. Recommendation

Change nothing in the frozen Stack v2 spec. Specifically:

- **Do NOT** filter entries by gamma levels, level proximity, headwind, gamma condition, or QScores — every such cut was null

or refuted on 81 days.

- **Do NOT** adopt "never trade into a level" — it was the wrong-direction rule in this window; clear-air trades were the worst bucket.
- **Keep capturing MenthorQ pre-market daily** (the app's Refresh button). The backfilled archive is EOD-stamped; only same-morning captures make the regime scalars trustworthy.
- **Three pre-registered re-tests at ~6–12 months of clean data** (one test each, no threshold tuning):

1. **Cluster skip/downsize** — stacked trades with a ≥ 3 -level cluster within 1R underperform (this study: +0.005 vs +0.186).
2. **D1-causal sizing** — mornings labeled negative-gamma realize more than implied → a sizing rule, not an entry rule.
3. **Seasonality_score ≥ 2 skip** — the -0.42R pocket (thinnest evidence of the three).

7. Caveats & open questions

- **One regime:** Mar–Jul 2026 (a vol spike into a long positive-gamma grind). 81 days cannot rule out effects smaller than $\sim \pm 0.2R$ per subgroup; nulls here are "not detectable at this n", not proofs of absence.
- **EOD-stamp contamination:** as-published gamma_condition/QScores may embed same-day info; D1's headline number is partially circular. The causal variants are the honest ones and are all n.s. — direction-consistent but unproven.
- **Multiplicity:** ~50 cuts total across four rounds; isolated [+] marks are expected by chance. Only pre-registered-direction + cross-scope-consistent findings (clusters, D1) were promoted to watch items.
- **Stage-1 scale:** the touch test used one fixed scale (6 bars / 3 pts). A different horizon might behave differently — but the containment test (day horizon) and first-touch test (EOD horizon) covered the coarser scales and agreed.
- Break-and-retest remains **untested** (event rate too low), not refuted.

8. Reproduce

- Data: data/menthorq/menthorq_levels.csv (+ raw JSON per day in data/menthorq/raw/), backfill: scripts/menthorq_backfill.py
- Round 1 (stages 0–2): scripts/menthorq_edge_study.py → full tables in docs/living/menthorq_edge_study_20260704_tables.md; sim cache data/menthorq/_study_sim_results.parquet
- Round 2 (containment/first-touch): scripts/menthorq_sr_followup.py → docs/living/menthorq_sr_followup_20260704.md
- Round 3 (user rules + universe): scripts/menthorq_userrules.py → docs/living/menthorq_userrules_20260704.md
- Round 4 (day regime): scripts/menthorq_dayregime.py → docs/living/menthorq_dayregime_20260704.md
- Signals: MyMicroChannel export ES SEP26 5M (Desktop, 1850 days); bars data/bars/_continuous.parquet; ticks data/ticks_continuous/; exec = S53 frozen spec via simulation_engine.simulate_trades + stack_filter.py.